

Task 2: API Integration & JSON Handling

```
def fetch_users():
    try:
        print("\nConnecting to API...\n")
        # API URL
        url = "https://jsonplaceholder.typicode.com/users"
        # Sending GET request
        response = requests.get(url, timeout=5)
        # Check for HTTP errors
        response.raise_for_status()
        # Convert JSON response into Python list
        users = response.json()
        print("API Data Fetched Successfully!\n")
        return users
    except requests.exceptions.HTTPError as http_error:
        print("HTTP Error Occurred:", http_error)
    except requests.exceptions.ConnectionError:
        print("Connection Error! Please check internet connection.")
    except requests.exceptions.Timeout:
        print("Request Timed Out!")
    except requests.exceptions.RequestException as error:
        print("Some Error Occurred:", error)
    return []

# -----
# Function to display all users
# -----

def display_users(users):
    if not users:
        print("No data available.")
        return
    print("\n===== ALL USERS =====\n")
    for user in users:
        print(f"ID      : {user['id']}")
        print(f"Name    : {user['name']}")
        print(f"Username : {user['username']}")
```

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print(f"Email   : {user['email']}")
print(f"City    : {user['address']['city']}")
print("-" * 40)
# -----
# Function to search user by name
# -----

def search_user(users):
    if not users:
        print("No data available for searching.")
        return

    search_name = input("\nEnter user name to search: ").lower()

    found = False

    print("\n===== SEARCH RESULT =====\n")

    for user in users:
        # Convert name to lowercase for case-insensitive search
        if search_name in user['name'].lower():
            found = True

            print(f"ID       : {user['id']}")
            print(f"Name     : {user['name']}")
            print(f"Username : {user['username']}")
            print(f"Email    : {user['email']}")
            print(f"Phone    : {user['phone']}")
            print(f"Website  : {user['website']}")
            print(f"Company  : {user['company']['name']}")
            print(f"City     : {user['address']['city']}")
            print("-" * 40)

            if not found:
                print("No matching user found.")

    # -----
    # Function to filter users by city
    # -----

    def filter_by_city(users):
        if not users:
            print("No data available.")
            return

```

```

city_name = input("\nEnter city name: ").lower()

found = False

print("\n===== USERS FROM CITY =====\n")

for user in users:

if city_name == user['address']['city'].lower():

found = True

print(f"Name : {user['name']}")

print(f"Email : {user['email']}")

print(f"Phone : {user['phone']}")

print("-" * 40)

if not found:

print("No users found from this city.")

# -----

# Main Program

# -----

def main():

# Fetch API data once

users = fetch_users()

while True:

print("\n===== MENU =====")

print("1. Display All Users")

print("2. Search User by Name")

print("3. Filter Users by City")

print("4. Exit")

choice = input("\nEnter your choice: ")

if choice == "1":

display_users(users)

elif choice == "2":

search_user(users)

elif choice == "3":

filter_by_city(users)

elif choice == "4":

print("\nProgram Exited Successfully!")

break

else:

```

```
print("\nInvalid Choice! Please try again.")
```

```
# -----
```

```
# Program Execution Starts Here
```

```
# -----
```

```
if __name__ == "__main__":
```

```
    main()
```

```
Connecting to API...

API Data Fetched Successfully!

===== MENU =====
1. Display All Users
2. Search User by Name
3. Filter Users by City
4. Exit

Enter your choice: 1

===== ALL USERS =====

ID      : 1
Name     : Leanne Graham
Username : Bret
Email    : Sincere@april.biz
City     : Gwenborough
-----
ID      : 2
Name     : Ervin Howell
Username : Antonette
Email    : Shanna@melissa.tv
City     : Wisokyburgh
-----
ID      : 3
Name     : Clementine Bauch
Username : Samantha
Email    : Nathan@yesenia.net
City     : McKenziehaven
-----
ID      : 4
Name     : Patricia Lebsack
Username : Karianne
Email    : Julianne.OConner@kory.org
City     : South Elvis
-----
ID      : 5
Name     : Chelsey Dietrich
Username : Kamren
Email    : Lucio_Hettinger@annie.ca
City     : Roscoeview
-----
ID      : 6
Name     : Mrs. Dennis Schulist
Username : Leopoldo_Corkery
Email    : Karley_Dach@jasper.info
City     : South Christy
-----
ID      : 7
Name     : Kurtis Weissnat
Username : Elwyn.Skiles
Email    : Telly.Hoeger@billy.biz
City     : Howemouth
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ID : 8
Name : Nicholas Runolfsdottir V
Username : Maxime_Nienow
Email : Sherwood@rosamond.me
City : Aliyaview

ID : 9
Name : Glenna Reichert
Username : Delphine
Email : Chaim_McDermott@dana.io
City : Bartholomebury

ID : 10
Name : Clementina DuBuque
Username : Moriah.Stanton
Email : Rey.Padberg@karina.biz
City : Lebsackbury

===== MENU =====

1. Display All Users
2. Search User by Name
3. Filter Users by City
4. Exit

Enter your choice: